

Reactivity of mixed organozinc and organocopper reagents. 3. Atom economic electrophilic amination of methyl arylzinc reagents

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Reaction of methyl arylzincs with acetone *O*-(mesitylenesulfonyl)oxime in THF in the presence of CuCN at room temperature is efficient in selective electrophilic amination of aryl carbanions. This procedure allows for the preparation of arylamines in moderate to good yields and provides an efficient and atom economic alternative to existing amination methods for diarylzincs. Copyright © 2009 John Wiley & Sons, Ltd.

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Introduction

Amines are synthetic and biologically important compounds.^[1] For arylamine synthesis,^[2,3] Cu-catalyzed Ullmann type amination,^[4] Pd-catalyzed Hartwig–Buchwald type amination^[5,6] and Ni-catalyzed amination of aryl halides with amines provide convenient methods.^[7] Electrophilic amination is also an important and potentially valuable methodology and an electrophilic amino group is transferred to carbon nucleophiles.^[8–14] A number of electrophilic aminating reagents, which are synthetic equivalents of NH_2^+ or NR_2^+ synthons, have been developed for electrophilic amination of ordinary carbanions^[9–11,14] and also for α -carbanions derived from carbonyl compounds.^[12–14] Electrophilic amination has been successfully used for the amination of aryllithium, -magnesium, -copper and -zinc reagents.^[2,8,11,12,14] Considering their higher reactivity, ease of preparation^[15–19] and tolerance to electrophilic functional groups,^[15–17] amination of diarylzincs provides an easy access to one-pot preparation of unfunctionalized and functionalized arylamines. Diarylzincs have been shown to react with aminating reagents such as *O*-methylhydroxylamine,^[14,20] *N*-mono- and *N,N*-disubstituted *O*-benzoylhydroxylamines,^[14,21] oxaziridines^[22] and arylazosulfonates^[23] in good yields.

We developed acetone *O*-(mesitylenesulfonyl)oxime **1**^[24,25] to aminate Grignard reagents and it was also successfully used for the amination of diarylzincs^[20] as well as arylzinc halides^[20,26] and triarylzincates^[20] under CuCN catalysis. Amination of diorganozincs and triorganozincates was found to give higher yields than that of organozinc chlorides.^[20]

However, in the reactions of diorganozincs, R_2Zn , and triorganozincates, R_3ZnMgBr , only one of the R groups is transferred to the electrophile, essentially wasting the other R groups. Mixed diorganozincs, $\text{R}^1\text{R}^2\text{Zn}$, in which one of the R^1 and R^2 groups has a lower rate of transfer than the other group, have been developed^[27] and ethyl, iso-propyl and *t*-butyl groups were found to react much more slowly than the other alkyl, vinyl and phenyl groups. Bolm used combinations of dialkylzinc and diarylzinc for aryl transfer^[28a–c] and this method was successfully applied.^[28d,e]

Mixed diorganozincs, $\text{R}_\text{R}\text{R}_\text{T}\text{Zn}$ have one transferable group R_T and one residual group, R_R .^[29] Knochel reported trimethylsilylmethyl, Me_3SiCH_2 ,^[29a] neopentyl, *t*-BuCH₂ and neophyl, $\text{PhMe}_2\text{CCH}_2$ groups^[29b] to be excellent residual groups. Mixed diorganozincs have been used in substitution reactions^[27f,29i,j], in 1,2-addition to aldehydes^[27a–d,27g–j,28,29a–c] and to imines^[29a] and in 1,4-addition to enones.^[27e,29c–h]

Using mixed diorganozincs provides an alternative to diorganozincs, which are potentially expensive and difficult to prepare. Our continuing work on the electrophilic amination^[30] and also on the transfer selectivity of the organyl groups in mixed diorganozincs,^[31] prompted us to explore the synthetic applicability of electrophilic amination of mixed diorganozincs.

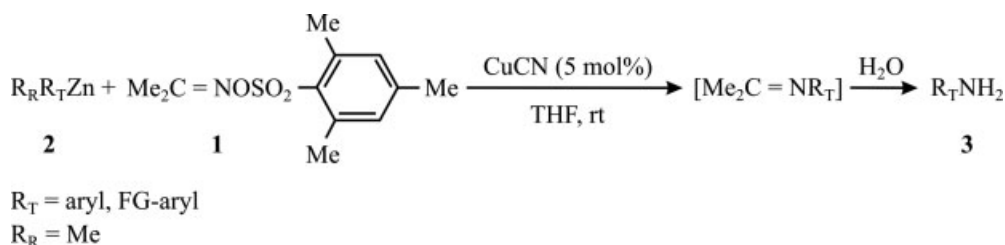
Herein, we report our findings for the development of a new method for $\text{C}_{\text{aryl}}-\text{N}$ coupling using methyl arylzincs and acetone *O*-(mesitylenesulfonyl)oxime **1** (Scheme 1). This method provides an efficient economic alternative to the amination of diarylzincs for arylamine synthesis.

Results and Discussion

Initially, we carried out initial experiments to find the reactivity of a mixed arylzinc in amination reaction with acetone *O*-(mesitylenesulfonyl)oxime **1**. For the residual group in mixed diarylzincs, we chose methyl group, depending on its previous successful use^[27g–j,28] in mixed alkylarylzincs as well as the commercial availability of methylmagnesium chloride or methyllithium to use as a methyl carbanion source. We prepared methylphenylzinc **2a** by allowing methylmagnesium chloride with phenylzinc

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Scheme 1. Electrophilic amination of methyl arylzincs **2** with acetone *O*-(mesitylenesulfonyl)oxime **1**.

chloride prepared by transmetalation of phenylmagnesium bromide. We tried amination of **2a** in THF under CuCN catalysis. As we had already succeeded in controlling organyl group transfer using *N*- or *O*-donor cosolvents in acylation of *n*-butylphenylzinc,^[31a,b] we screened NMP, HMPA, TMEDA and diglyme as cosolvents in the amination of **2a**. However, amination yield of **2a** was found to be lower in THF:cosolvent mixtures. From further experiments, the optimized yield was found to be 65% by stirring 1.5 equiv. of **2a** with 1 equiv. of aminating reagent **1** in THF in the presence of 5 mol% CuCN at room temperature for 1 h. Considering the amination yield of 51% of Ph₂Zn,^[20b] it is obvious that PhMeZn **2a** could be aminated with a higher yield. This study also revealed that the use of Me group instead of Ph group as residual group in Ph₂Zn increases the yield of phenyl transfer in diorganozinc reagent to electrophilic nitrogen. Further, we tested the generality of this amination protocol for methylarylzinc reagents. We had already obtained higher yields in the amination of Grignard reagent-derived organozincs,^[20] supporting the role of MgCl₂ as a Lewis acid. Therefore, in the amination of methylarylzincs, we used magnesium-based organozinc reagents and we prepared methylarylzincs by the *in situ* method,^[27g,29a,b,31a,b] i.e. we reacted arylmagnesium bromide and methylmagnesium chloride with ZnCl₂ (methods A and B).

Method A: phenylmagnesium bromide **2a** and substituted phenylmagnesium bromides, FG-C₆H₄MgBr **2b–f** (FG = 4-Me **b**, 3-Me **c**, 4-MeO **d**, 3-MeO **e**, 4-Br **f**) (1 equiv.) were prepared by direct insertion of magnesium, then transmetalated to FG-C₆H₄ZnCl with ZnCl₂ (1 equiv.) and reacted with MeMgCl (1 equiv.).

Method B: substituted phenylmagnesium bromides, FG-C₆H₄MgBr **2g–i** (FG = 4-MeOOC **g**, 4-NC **h**, 4-MeOC **i**) (1 equiv.) were prepared by I/Mg exchange from requisite aryl iodide and isopropylmagnesium bromide^[19] and then reacted with MeZnCl (1 equiv.) prepared by transmetalation of MeMgCl with ZnCl₂.

A broad range of methylarylzincs reacted with aminating reagent **1** at room temperature within 1 h in moderate to good yields. The results are summarized in Table 1. The data are averages of at least two independent experiments. The mildness of the conditions involved allows a certain tolerance for mixed arylzincs bearing electron-drawing substituents as well electron-withdrawing ones. Unfortunately, all our efforts failed to aminate of mixed phenylzinc bearing 4-MeOC-group.

Conclusion

In summary, reaction of mixed methylarylzincs with acetone *O*-(mesitylenesulfonyl)oxime in THF in the presence of CuCN

Table 1. CuCN catalyzed amination of methylarylzinc reagents **2** with acetone *O*-(mesitylenesulfonyl)oxime

Entry	R	Yield, % ^c
1	C ₆ H ₅ a	66
2	4-MeC ₆ H ₄ b	49
3	3-MeC ₆ H ₄ c	48
4	4-MeOC ₆ H ₄ d	50
5	3-MeOC ₆ H ₄ e	38
6	4-BrC ₆ H ₄ f	48
7	4-MeOOC ₆ H ₄ g	45
8	3-NCC ₆ H ₄ h	34

^a **2a–f** were prepared *in situ* by reacting RZnCl with MeMgCl in THF. RZnCl was prepared by transmetalation of RMgBr prepared by direct insertion of Mg (method A). See Experimental section. **2g, h** were prepared *in situ* by reacting RMgBr with MeZnCl in THF. RMgBr was prepared by reacting RI with *i*-PrMgBr in THF (method B). See Experimental section.

^b Reactions were carried out on 5 mmol aminating reagent **1**, which was added to a mixture of RZnMe **2** and CuCN (5 mol%) in THF (RZnMe:**2** molar ratio is 1.5)

^c Yield of amine **3** isolated as *N*-benzoyl derivative.

at room temperature is efficient in selective amination of aryl carbanions. With applicability, good yields and low cost, this procedure provides an atom economic alternative to existing amination methods of diarylzincs for arylamine synthesis. Further investigation for the amination of mixed organozinc reagents is needed. We are currently examining ways to extend the utility of electrophilic amination.

Experimental

All reactions were carried out in oven-dried glassware under a nitrogen atmosphere using standard syringe-septum cap techniques.^[32] GC analyses were performed on a Thermo-Finnigan gas chromatograph equipped with a ZB-5 capillary column packed with phenyl polysiloxane using the internal standard technique. ¹H NMR spectra were obtained using a Varian Mercury 400 operating at 400 MHz. THF was distilled from sodium benzophenone dianion. Pure NMP, HMPA, TMEDA were distilled before use. Commercially available aryl bromides and iodides were used without further purification. ZnCl₂ was dried under reduced pressure at 100 °C for 2 h and dissolved in THF just before use. CuCN was purified according to the published procedure.^[33] Arylmagnesium

bromides were prepared in THF by standard method or Knochel's I/Mg exchange method.^[19]

Typical Procedure for the CuCN-catalyzed Amination of Methyl Arylzincs, RMeZn **2** with Acetone *O*-(mesitylenesulfonyl) oxime **1**

Preparation of RMeZn reagents **2a–f** (method A)

First RMgBr (R = C₆H₅ **a**, 4-MeC₆H₄ **b**, 3-MeC₆H₄ **c**, 4-MeOC₆H₄ **d**, 3-MeOC₆H₄ **e**, 4-BrC₆H₄ **f**) reagent was prepared by direct insertion of Mg in THF. A flame-dried, two-necked flask equipped with a stirring bar and maintained under nitrogen atmosphere at –20 °C was charged with ZnCl₂ solution in THF (11.8 ml, 0.64 M in THF, 7.5 mmol). RMgBr solution in THF (7.5 mmol) and then MeMgCl solution in THF (7.5 mmol) was added dropwise and the mixture was stirred at this temperature for 15 min.

Preparation of RMeZn reagents **2** (method B)

First RMgBr reagent (R = 4-MeOCC₆H₄ **g**, 3-NCC₆H₄ **h**, 4-MeOCC₆H₄ **i**) (7.5 mmol) was prepared by reacting RI (7.5 mmol) with *i*-PrMgBr (7.5 mmol) in THF at –20 °C. Another flask was charged with ZnCl₂ solution in THF (7.5 mmol) at –20 °C and first MeMgCl solution in THF (7.5 mmol) and then freshly prepared RMgBr solution in THF was added dropwise and the mixture was stirred at this temperature for 15 min.

Amination of methyl arylzincs

For amination to RMeZn **2** solution at –20 °C, CuCN (0.033 g, 0.375 mmol) and acetone *O*-(mesitylenesulfonyl)oxime **1** solution in THF (1.280 g, 5 mmol in 13 ml THF) were added. The temperature was allowed to rise to room temperature and stirring at this temperature was continued for 1 h. For hydrolytic work-up, the mixture was stirred with concentrated HCl (15 ml) for 2 h. The aqueous phase was washed with Et₂O, made basic with concentrated NaOH and free amine was extracted with Et₂O. The organic layer was dried over Na₂SO₄. The crude amines were obtained after evaporation of solvent under reduced pressure. For convenience in isolation, the crude amines **3** were converted to their *N*-benzoyl derivatives by reaction with benzoyl chloride in the presence of NaOH.^[20b,34] Crude derivatives were recrystallized from ethanol or ethanol–water as solids and their melting points^[35] and ¹H NMR spectra are consistent with those reported in the literature.^[20b,35]

Acknowledgments

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